Vertex formula

1 Consider the parabola of the form $y = ax^2 + bx + c$, where $a \neq 0$.

a What does the line $x = \dfrac{-b}{2a}$ represent on the parabola?

b Explain how to find the coordinates of the vertex using a, b and c.

2 Find the coordinates of the vertex of $y = 3x^2 - 6x - 9$.

3 Find the maximum value of y for the quadratic function $y = -x^2 + 10x - 25$.

4 Find the x-coordinate of the vertex of the parabola represented by $P(x) = px^2 - \dfrac{1}{2}px - q$.

Vertex form

5 Consider two different parabolas:

- Parabola 1 has two x-intercepts at $(-6, 0)$ and $(4, 0)$.
- Parabola 2 has a vertex at $(-1, 3)$.

a How many unique parabolas have the same intercepts as Parabola 1?

b How many unique parabolas have the same vertex as Parabola 2?

c How many unique parabolas have the intercepts as Parabola 1 and the same vertex as Parabola 2?

d The equation of the parabola from part (c) can be written as $y = a(x + 6)(x - 4)$. Find the value of a and write the equation of the parabola in standard form.

6 Consider the parabola defined by the equation $y = a(x - h)^2 + k$.

a What does the point (h, k) represent.

b What is the least number of points required to graph the parabola.

7 Write $y = x^2 + 10x - 23$ in vertex form.

8 Consider the equation $y = x^2 + 4x$.

a Write the equation in vertex form.

b Find the coordinates of the vertex.

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9 Consider the parabola $y = 8 + 3(x - 7)^2$:

a Find the coordinates of the vertex of the parabola.

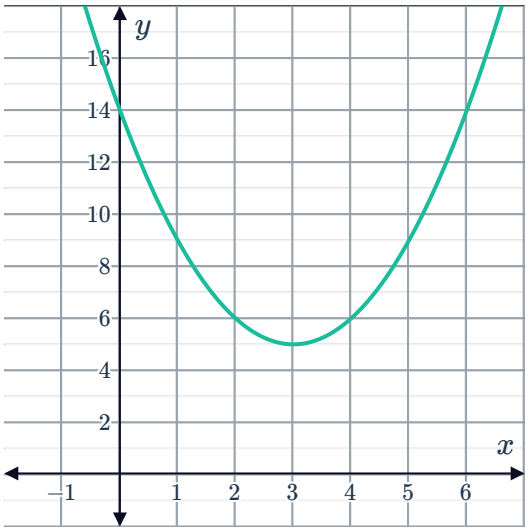
b The parabola is reflected about the y-axis. Find the new equation of the parabola formed after the reflection.

c Find the coordinates of the vertex of the parabola formed after the reflection.

10 Consider the following graph:

a Find the coordinates of the vertex.

b Find the equation of the parabola.



Graphs of quadratics

11 For each quadratic function:

i Find the x-intercepts.

ii Find the y-intercept.

iii Find the axis of symmetry.

iv Find the coordinates of the vertex.

v Sketch the graph.

a $y = x^2 + 4x$

b $y = x^2 + 2x - 8$

c $y = 2x^2 + 9x + 9$

d $y = -2x^2 + 16x - 24$

12 For each quadratic function:

i Find the x-intercepts.

ii Find the y-intercept.

iii Find the coordinates of the vertex.

iv Sketch the graph.

a $y = (x + 1)^2$

b $y = (x - 3)^2 - 1$

c $y = -(x + 2)^2 + 4$

d $y = 9 - (x - 5)^2$

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13 For each quadratic function:

i Find the axis of symmetry.

ii Find the coordinates of the vertex.

iii Find the y-intercept.

iv Sketch the graph.

a $y = x^2 - 4x + 8$

b $y = -x^2 - 4x - 9$

c $y = 2\left(x - \frac{3}{4}\right)^2 - \frac{1}{4}$

14 For each quadratic function:

i Find the coordinates of the vertex.

ii Sketch the graph.

a $P(x) = x^2 - 4x + 2$

b $P(x) = x^2 + 5x + 3$

c $P(x) = -x^2 - 4x + 3$

d $P(x) = 2x^2 - 8x + 4$

e $P(x) = -2x^2 - 8x + 2$

f $P(x) = -2x^2 - 10x - 8$

15 A parabola of the form $y = (x - h)^2 + k$ is symmetrical about the line $x = 2$, and its vertex lies 6 units below the x-axis.

a Write the equation of the parabola.

b Sketch the graph.

16 For each of the following functions:

i Rewrite the function in the form $y = (x + a)^2 + b$.

ii Find the coordinates of the vertex.

iii Sketch the graph.

a $y = x^2 - 6x + 1$

b $y = x^2 + 6x$

17 Consider the quadratic $y = x^2 - 12x + 32$

a Find the zeros of the quadratic function.

b Express the equation in the form $y = a(x - h)^2 + k$.

c Find the coordinates of the vertex.

d Sketch the graph of the parabola.

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18 For each quadratic function:

i Find the coordinates of the vertex.

ii Find the axis of symmetry.

iii Sketch the graph.

iv Sketch the axis of symmetry on the same axes.

a $f(x) = (x-3)^2$ **b** $h(x) = (x+5)^2$

c $f(x) = -3(x+2)^2 - 4$

19 Sketch the graph of the following functions:

a $y = (x-2)^2$ **b** $y = 2(x-2)^2$ **c** $y = 2(x-2)^2 + 2$

Applications

20 When an object is thrown into the air, its height above the ground is given by the equation

$$h = 193 + 24s - s^2$$

where s is its horizontal distance from where it was thrown.

a Find the value of s, at the point when it reaches its greatest height above the ground.

b Find the maximum height reached by the object.

21 The height at time t of a ball thrown upwards is given by the equation $h = 59 + 42t - 7t^2$.

a How long does it take the ball to reach its maximum height?

b Find the maximum height reached by the ball.

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